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SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**
Course Code: **MLA02**
Course Name: **FUNDAMENTALS OF MACHINE LEARNING**
Duration: **1:30 Hr**
Date : _____

46/50 +

Max Marks: **50 Marks**

ASSIGNMENT – I

UNIT I – INTRODUCTION AND MATHEMATICAL FOUNDATIONS

Topics: *Design of a learning system, types of machine learning, probability theory and distributions, Bayes decision theory, information theory*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO1	Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory.

Answer all the Tasks

Task 1. A retail bank plans five machine-learning systems. For each system, analyse the problem and state (i) the learning paradigm (supervised / unsupervised / semi-supervised / reinforcement), (ii) the learning task (classification, regression, clustering, control, etc.) and (iii) the experience E and performance measure P in Mitchell's (T, P, E) formulation:

(a) Predict whether a loan applicant will default, using ten years of labelled repayment records.

(b) Discover behavioural customer segments among 2 million customers for which no labels exist.

(c) Estimate the next-month credit-card spend (in ₹) of each customer.

(d) Learn an ATM cash-refill policy that maximises cash availability minus transport cost through repeated trial and feedback.

(e) Only 2% of 500,000 transaction records are labelled fraud/genuine; the rest are unlabelled. Build a fraud detector that exploits both kinds of data.

Finally, justify which of the five systems is most harmed by label noise.

(5) (CO1) (BL4)

Task 2. A screening test for a disease with prevalence 2% has sensitivity 95% and specificity 90%. Calculate the probability that a person who tests positive actually has the disease, and analyse why the answer is much lower than the sensitivity.

(2) (CO1) (BL4)

Task 3. The joint probability mass function of a discrete random vector (X, Y) is given below.

X \ Y	Y = 0	Y = 1
X = 0	0.10	0.20
X = 1	0.15	0.25

X \ Y	Y = 0	Y = 1
X = 2	0.20	0.10

Analyse the distribution by (a) finding the marginals of X and Y, (b) computing $E[X]$, $Var(X)$ and $E[Y]$, (c) computing $P(X = 2 \mid Y = 1)$, (d) computing $Cov(X, Y)$ and the correlation coefficient, and (e) deciding whether X and Y are independent, with a justification.

(6) (CO1) (BL4)

Task 4. A temperature sensor output is modelled as $X \sim N(\mu = 50, \sigma^2 = 16)$.

(a) Calculate $P(X > 58)$ and $P(46 < X < 54)$.

(b) A second sensor gives the readings 12, 15, 11, 14, 13. Derive the maximum-likelihood estimates of its mean and variance and compare the ML variance with the unbiased estimate, stating which is biased and why. (Use $\Phi(1) = 0.8413$, $\Phi(2) = 0.9772$.)

(4) (CO1) (BL4)

Task 5. A source emits four symbols with probabilities (1/2, 1/4, 1/8, 1/8). Calculate its entropy, compare it with the entropy of a uniform source over four symbols, and analyse the redundancy. Also compute the average length of the code {0, 10, 110, 111} and comment on its optimality.

(3) (CO1) (BL4)

Task 6. A model predicts class probabilities Q = (1/3, 1/3, 1/3) while the true distribution is P = (0.5, 0.25, 0.25). Compute $KL(P||Q)$ and $KL(Q||P)$ in bits, and analyse what the asymmetry implies when KL divergence is used as a training loss.

(3) (CO1) (BL4)

Task 7. Two binary variables A and B have the joint distribution $P(0,0) = 0.4$, $P(0,1) = 0.1$, $P(1,0) = 0.1$, $P(1,1) = 0.4$. Calculate H(A), H(B), the joint entropy H(A,B), the conditional entropy H(A|B) and the mutual information I(A;B). Analyse what the value of I(A;B) says about using B as a feature to predict A.

(4) (CO1) (BL4)

Task 8. A one-dimensional feature x follows $p(x|\omega_1) = N(2, 4)$ for class ω_1 and $p(x|\omega_2) = N(6, 4)$ for class ω_2, with priors $P(\omega_1) = 0.6$ and $P(\omega_2) = 0.4$ and zero–one loss. Derive the Bayes decision threshold x^*, classify $x = 4.5$, and calculate the overall probability of error. Analyse how the threshold moves as the prior of ω_1 increases.

(4) (CO1) (BL4)

Task 9. For an observation x, the posteriors are $P(\omega_1|x) = 0.3$ (disease) and $P(\omega_2|x) = 0.7$ (healthy). Action α_1 = 'declare disease', α_2 = 'declare healthy'. The loss matrix is $\lambda(\alpha_1|\omega_1) = 0$, $\lambda(\alpha_1|\omega_2) = 8$ (unnecessary treatment), $\lambda(\alpha_2|\omega_1) = 50$ (missed disease), $\lambda(\alpha_2|\omega_2) = 0$. Evaluate the conditional risks, decide the minimum-risk action, compare it with the MAP decision and determine the posterior $P(\omega_1|x)$ above which α_1 is optimal.

(3) (CO1) (BL5)

Task 10. Polynomial regression models of increasing degree were fitted to the same dataset, giving the mean-squared errors below.

Degree	1	2	3	5	9
Training MSE	9.0	4.1	2.2	0.9	0.1

Degree	1	2	3	5	9
Validation MSE	9.5	4.6	2.9	3.6	11.8

Calculate the generalisation gap for each degree, diagnose under-fitting/over-fitting using the bias–variance trade-off, select the best model with justification, and recommend two techniques to improve the degree-9 model.

(4) (CO1) (BL5)

Task 11. A coin is tossed 10 times and shows 7 heads. The bias θ = P(head) is given a Beta(2, 2) prior. Calculate the ML estimate, the MAP estimate and the posterior mean of θ. Then evaluate how the three estimates change if the data were 70 heads in 100 tosses, and explain the role of the prior.

(4) (CO1) (BL5)

Task 12. A hospital wants an early-warning system that raises an alert when an ICU patient is likely to develop sepsis within the next 6 hours. Historical electronic health records (vital signs, lab values, outcomes) are available.

(a) Design the learning system: state the task T, performance measure P and experience E, the target function, a suitable input representation, a learning algorithm and the validation strategy. [3]

(b) On a validation cohort of 1,000 patients, 5% develop sepsis. The trained model has sensitivity 0.90 and specificity 0.80. Compute TP, FN, FP and TN. [1]

(c) Compute the positive predictive value (PPV) and negative predictive value (NPV). [1]

(d) A missed sepsis case costs ₹100 (unit cost) and a raise alarm costs ₹5. Compare the total cost of (i) using the model, (ii) alerting every patient and (iii) never alerting. [2]

(e) Recommend a deployment policy and justify how the alert threshold should be set using Bayes decision theory. [1]

(8) (CO1) (BL6)

ANSWER SCRIPT EVALUATION SHEET
ASSIGNMENT - I

Programme	B.E / B.Tech	Course Code	MLA02
Course Name	Fundamentals of Machine learning.	Date	
Duration	1 Hr 30 Min	Max Marks	50

Register No.: 1911260190

TASK	Max Marks	Marks Awarded
Task 1	5	
Task 2	2	
Task 3	6	
Task 4	4	
Task 5	3	
Task 6	3	
Task 7	4	
Task 8	4	
Task 9	3	
Task 10	4	
Task 11	4	
Task 12	8	
TOTAL	50	

Total in words:

Assignment-1
Unit-1

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1. An: T(Task): what the system has to learn or perform.

P(Performance): How the system's performance is measured

E(Experience): The data or experience used for learning

Case	Learning paradigm	Learning Task
loan default	Supervised learning	Classification
Customer segmentation	Unsupervised learning	Clustering
ATM cash refill	Reinforcement learning	Control and decision making

Explanation:

a) loan default prediction: The model learns from historical labelled records to predict whether a customer will default or repay a loan.

b. Customer segmentation: The model group customers on similar spending and behavioral pattern without predefined labels.

c. Credit-card spending: the model predicts the next month credit-card spending in rupees.

d. ATM cash-refill policy: The system learns the best time and quantity for refilling cash through rewards and penalties.

e. Fraud detection: The model learns from small amount of labelled transaction data.

Conclusion:

By Different ML problem require different learning paradigm Selection the correct paradigm.

Given:

Disease prevalence = 2% = 0.02

Sensitivity = 95% = 0.95

Specificity = 90% = 0.90

False positive rate = 10% = 0.1

Formula:

$$P(D|+) = \frac{P(+|D)\,P(D)}{P(+)}$$

$$= \frac{0.95 \times 0.02}{0.95 \times 0.02 + 0.1 \times 0.98}$$

$$= \frac{0.019}{0.117}$$

$$\approx 0.1624$$

∴ The probability of actually having the disease after a positive result is 16.24%.

3. Given:

b.

X	y = 0	y = 1
0	0.1	0.2
1	0.15	0.25
2	0.2	0.1

c.

Marginal probabilities of x and y

c

$$P(X=x) = \sum_{y} P(X=x, Y=y)$$

$$P(X=0) = 0.10 + 0.20 = 0.30$$

$$P(X=1) = 0.15 + 0.25 = 0.40$$

$$P(X=2) = 0.20 + 0.10 = 0.30$$

Margin probabilities of y:

$$P(Y=0) = 0.10 + 0.15 + 0.20 = 0.45$$

$$P(Y=1) = 0.20 + 0.25 + 0.10 = 0.55$$

Expected value of x

$$E|X| = \sum x P(x)$$

$$= (0)(0.30) + (1)(0.40) + 2(0.30)$$

$$E|X| = 1$$

variance of X

$$\text{Var}(X) = E[x^2] - [E[X]]^2$$

$$E[x^2] = 0^2(0.30) + 1^2(0.40) + 2^2(0.30)$$

$$= 1.60$$

$$\text{Var}(X) = 1.60 - 1^2$$

$$= 0.60$$

Expected value of Y

$$E[y] = \sum y P(y)$$

$$= (0)(0.45) + (1)(0.55)$$

$$E[Y] = 0.55$$

conditional probability

$$P(X=2|Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)}$$

$$= \frac{0.10}{0.55}$$

$$P(X=2|Y=1) = 0.1818$$

covarion and correlation

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

$$E[XY] = 1 \times 1 \times 0.25 + 2 \times 1 \times 0.1$$

$$= 0.45$$

$$Cov(X, Y) = 0.45 - 1(0.5)$$

$$Cov(X, Y) = -0.10$$

correlation coeffi

$$r = \frac{Cov(X, Y)}{\sqrt{Var(X) Var(Y)}}$$

$$= \frac{-0.10}{\sqrt{0.60 \times 0.14}}$$

$$r = -0.2593$$

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given:

$P(X > 58)$ and $P(46 < X < 54)$

Given:

$$X \sim N(\mu = 50, \sigma^2 = 16)$$

$$\mu = 50$$

$$\sigma = \sqrt{16} = 4$$

$$Z = \frac{X - \mu}{\sigma}$$

$P(X > 58)$:

$$Z = \frac{58 - 50}{4} = 2$$

$$\phi(2) = 0.9772$$

$$P(X > 58) = 1 - 0.9772 = 0.0228$$

$P(46 < x < 54)$:

$x = 46$:

$$z_1 = \frac{46 - 50}{4} = -1$$

$x = 54$

$$z_2 = \frac{54 - 50}{4} = 1$$

$$\phi(1) - \phi(-1)$$

$$= 0.841 - 0.15$$

$P(46 < x < 54) = 0.6926$

b. Given:

12, 15, 11, 14, 13

$$\hat{\mu} = \frac{\sum x_i}{n}$$

$$= \frac{65}{5} = 13$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \hat{\mu})^2$$

$$= \frac{1 + 4 + 4 + 1 + 0}{5}$$

$$\sigma^2 = 2$$

5.

Given:

$$P = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right)$$

$$H(x) = -\sum P(x_i) \log_2 P(x_i)$$

$$H(x) = -\left[\frac{1}{2}\log_2 \frac{1}{2} + 0.25 \log_2 0.25\right]$$

$$= 1.75 \text{ bits}$$

$$H_{max} = \log_2 4 = 2 \text{ bits}$$

$$R = H_{max} - H$$

$$= 2 - 1.75 = 0.25 \text{ bits}$$

2

Percentage redundancy:

$$\frac{0.25}{2} \times 100 = 12.5\%$$

3 $$L = 0.5(1) + 0.25(2) + 0.125(3) + 0.125(3)$$

$$= 1.25 \text{ bits}$$

Given:

$$P = (0.5,\ 0.25,\ 0.25)$$

$$Q = (1/3,\ 1/3,\ 1/3)$$

$$D_{KL}(P\|Q) = \sum P(x) \log_2 \frac{P(x)}{Q(x)}$$

$$= 0.5 \log_2 (1.5) + 0.25 \log_2 (0.75)$$

$$D_{KL}(P\|Q) = 0.0850 \text{ bits}$$

$$D_{KL}(Q\|P) = \sum Q(x) \log_2 \frac{Q(x)}{P(x)}$$

$$= \frac{1}{3}\left[\log_2 (2/3) + 2 \log_2 (4/3)\right]$$

$$D_{KL}(Q\|P) = 0.087 \text{ bits}$$

$$\therefore D_{KL}(P\|Q) \neq D_{KL}(Q\|P)$$

7.

Given:

Marginal probabilities:

$$P(A=0) = P(A=1) = 0.5 = P(B=0) = P$$

$$H(A) = -\sum P(A) \log_2 P(A)$$

$$= -\left[\frac{1}{2} \log_2 \frac{1}{2} + 0.5 \log_2 0.5\right)$$

HCF 1 bit

$$H(A) = 1 \text{ bit}$$

$$\therefore H(B) = 1 \text{ bit}$$

$$H(A,B) = -\sum P(A,B) \log_2 P(A,B)$$

$$= -[2(0.4) \log_2 0.4 + 2(0.1) \log_2 0.1$$

$$= 1.0575 + 0.6644$$

$$H(A,B) = 1.7219 \text{ bits}$$

$$H(A|B) = H(A,B) - H(B) = 1.7219 - 1 = 0.7219$$

$$I(A;B) = H(A) + H(B) - H(A,B)$$

$$= 1 + 1 - 1.7219$$

$$I(A;B) = 0.2781 \text{ bits}$$

$\therefore$ uncertainty about A is $\approx 27.8\%$.

$\mu_1 = 2$, $\mu_2 = 6$ $\sigma^2 = 4$, $P(\omega_1) = 0.6$, $P(\omega_2) = 0.4$

$$x^* = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma^2}{\mu_2 - \mu_1} \ln\left(\frac{P(\omega_1)}{P(\omega_2)}\right)$$

$$= \frac{2+6}{2} + \frac{4}{6-2} \ln \frac{0.6}{0.4}$$

$$x^* = 4.4055$$

$x = 4.5$

$4.5 > 4.4055$

$\therefore$ it belong to ω_2

Class 1:

$$P(\text{error}(\omega_1) = 1 - \phi\left(\frac{4.4055 - 2}{2}\right)$$

$$= 0.1145$$

$$P(\text{error}(\omega_2) = \phi\left(\frac{4.4055 - 6}{2}\right)$$

$$= 0.2126$$

$$P(\text{error}) = 0.6(0.1145) + 0.4(0.2126)$$

$$= 0.0687 + 0.0850$$

$$P(\text{error}) = 0.1537 \approx 15.4\%$$

9. given:

$$P(\omega_1|x) = 0.3$$

$$P(\omega_2|x) = 0.7$$

$$R(\alpha_1|x) = 0(0.3) + 8(0.7)$$

$$R(\alpha_1|x) = 5.6$$

$$R(\alpha_2|x) = 50(0.3) + 0$$

$$= 15$$

$$5.6 < 15$$

$\therefore$ min risk $= \alpha_1$

$$P = P(\omega_1|x)$$

$$8(1-P) = 50P$$

$$8 = 58P$$

$$P = 0.1379$$

$$\therefore P(\omega_1|x) > 0.138$$

tion:

Degree	Training MSE	Validation MSE
1	9.0	9.5
2	4.1	4.6
3	2.2	2.9
5	0.9	3.6
9	0.1	11.8

Generalisation Gap = Validation MSE − Training MSE

Degree	Calculation	Gap
1	9.5 − 9	0.5
2	4.6 − 4.1	0.5
3	2.9 − 2.2	0.7
5	3.6 − 0.9	2.7
9	11.8 − 0.1	11.7

Degree 1: Both error are high, indicating high bias and underfitting

Degree 9: Training error is very low, but validation error is high, indicating high

Degree 5: shows sign of overfitting because the generalisation gap is low

∴ Best model = Degree 3

Given:

Prior = Beta(2, 2)

number of toss = 10

H = 7, T = 3

$$\hat{\theta}_{ML} = \frac{h}{n}$$

$$= \frac{7}{10}$$

$$\hat{\theta}_{ML} = 0.700$$

Posterior distribution

Beta(h + a, n − h + b)

= Beta(7 + 2, 3 + 2)

= Beta(9, 5)

$$\hat{\theta}_{MAP} = \frac{h + a - 1}{n + a + b - 2} = \frac{7 + 2 - 1}{10 + 2 + 2 - 2}$$

$$= \frac{8}{12} = 0.6667$$

mon

$$E[\theta] = \frac{h+a}{n+a+b}$$

$$E[\theta] = \frac{7+2}{10+2+2} = \frac{9}{14} = 0.6429$$

2

sk (T):

edict whether an ICU patient will develop sepsis ithin the next 6 hrs. This is a binary classification roblem

erformance measure (P):

Sensitivity or recall

False - positive rate

AUC score

Expected misclassification cost

Experience (E):

Historical electronic health record containing patient vital sign, laboratory value

Target function:

$$f(x(t)) \rightarrow \{0, 1\}$$

c. +ve Predicted value

$$PPV = \frac{TP}{TP+FP} = \frac{45}{45+190} = 0.1915 = 19.1\%$$

-ve Predicted value

$$NPV = \frac{TN}{TN+FN} = \frac{760}{765} = 99.35\%$$

Cost of missed sepsis = ₹100

Cost of false alarm = ₹5

$$\text{Cost} = (FN \times 100) + (FP \times 5)$$

$$= (5 \times 100) + (190 \times 5)$$

$$= 500 + 950$$

Cost = ₹1450

Alerting every patient

$$\text{cost} = 950 \times 5$$

cost = ₹4750

$$P(\text{Sepsis} \mid x) > \frac{C_{FP}}{C_{FP} + C_{FN}}$$

$$= \frac{5}{5+100} = 0.0476$$

$$P(\text{Sepsis} \mid x) > 0.0476$$

∴ The alert threshold should be approximately 4.76 % because the cost of missing Sepsis is much high.

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